

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	
Project Name	AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Loan Account Management	Create loan account with lender, amount, EMI, overdue months, income Edit loan details Delete loan account
FR-2	Financial Health Analysis	Calculate EMI-to-income ratio and monthly surplus Calculate 0–100 debt-stress score and level Generate settlement percentage recommendation
FR-3	AI Negotiation Letter Generation	Generate settlement request letter Generate negotiation email Select tone: professional, firm, or empathetic
FR-4	Negotiation History Tracking	Persist every generated letter with its stress level and settlement % View past letters for a specific loan, most recent first
FR-5	Dashboard & Reporting	View aggregate totals across all loans (outstanding, EMI load) View count of critical-stress accounts
FR-6	Input Validation	Reject non-positive loan amounts, EMI, or income before reaching business logic Return clear validation error messages to the frontend

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, plain-English forms and clear visual indicators (stress gauge, color-coded badges) so non-technical borrowers can use the platform without training.
NFR-2	Security	The Gemini API key and all secrets are stored only in environment variables, never committed to source control; every backend request is validated via Pydantic schemas before reaching business logic.
NFR-3	Reliability	The financial-analysis engine returns results deterministically with no dependency on network availability; a template-based fallback ensures negotiation letters always generate even if the Gemini API is unavailable.
NFR-4	Performance	The dashboard summary endpoint returns all loan and analysis data in a single response, avoiding repeated N+1 API calls from the frontend.
NFR-5	Availability	Core loan management and financial analysis features remain fully functional with zero external API dependencies.
NFR-6	Scalability	Backend (Render) and frontend (Vercel) are independently deployable, allowing each tier to scale separately as usage grows.